

Evaluation of Bridges in the City of Omaha - Nebraska

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1 Executive Summary

1.1 Overview

This report examines the risk of bridge failure in Omaha, Nebraska over the next five years. Using 2025 federal bridge inspection data, 390 bridges in Douglas County were evaluated on two dimensions: how structurally sound they are, and how many people rely on them daily. The goal is to help decision-makers identify which bridges need attention most urgently, and why.

1.2 Methodology

Bridges were classified as higher-risk if any structural condition rating of deck, superstructure, substructure fell at or below 4, consistent with FHWA (Federal Highway Association) conventions. Descriptive statistics and Spearman correlation tests were used to examine relationships between bridge age, structural condition, and Average Daily Traffic (ADT), where ADT served as a proxy for the consequence of failure.

1.3 Key Findings

Of the 390 bridges examined, 16 (4.1%) were flagged as higher-risk. These bridges are on average 30 years older than the rest (mean age 69.8 vs. 40.0 years), and older bridges were consistently found to be in worse structural condition. Most bridges across the county are in reasonable shape, with average inspection scores falling in the good range.

One concern stands out: there is little relationship between how much traffic a bridge carries and how well it is maintained. Several structurally compromised bridges still carry tens of thousands of vehicles per day, representing the greatest risk to public safety and transportation.

1.4 Conclusion and Recommendations

Bridge risk in Omaha is concentrated in a small number of aging structures. The 16 higher-risk bridges should be prioritized for immediate inspection and repair — especially those that also carry high traffic volumes. More broadly, Omaha’s infrastructure management should move away from treating all bridges the same, and instead weigh both structural condition and public reliance together when deciding where to invest maintenance resources.

2 Introduction

In 2025, a report from the American Road and Transportation Builders Association (ARTBA) (2025) estimated that more than 40,000 bridges in the United States are structurally deficient, highlighting widespread concerns about aging infrastructure. Data from the Federal Highway Administration (2025) further show that a substantial share of bridges remain in fair or poor condition, requiring ongoing monitoring and maintenance. These statistics underscore the real risks associated with infrastructure deterioration, where failures can lead not only to economic disruption but also to severe safety consequences for the public.

While national reports emphasize the scale of the problem, they often rely on condition ratings alone, which do not fully capture the actual risk posed by individual bridges. In practice, bridges with similar structural conditions can present very different levels of danger depending on how heavily they are used. This creates a critical gap in infrastructure assessment, where decision-makers lack a clear method for prioritizing repairs based on both likelihood of failure and potential impact.

This report investigates bridge risk in Omaha (Douglas County) using a risk-based framework that evaluates safety as a function of both failure probability and consequence. Drawing on Federal Highway Administration (2025) element data, the analysis incorporates structural condition indicators (deck, superstructure, and substructure ratings, along with bridge age) and consequence measures such as average daily traffic. By examining how these factors interact, the study identifies bridges where deterioration coincides with high usage, representing the greatest overall risk.

Findings indicate that while most bridges are not at immediate risk of collapse, those with moderate structural degradation and high traffic exposure account for the highest-risk group. These results suggest that infrastructure management should shift from condition-based assessments toward risk-informed prioritization, enabling more effective allocation of maintenance resources and improved public safety outcomes.

3 Data & Method

3.1 Business Question

What is the risk of a bridge collapse in Omaha over the next five years, and what interventions should be implemented?

3.2 Research Question

Based on publicly available inspection and transportation data, what is the estimated probability of structural bridge failure in Omaha, and which bridges present the greatest risk when considering both likelihood and consequence?

3.3 Data & Methods

The analysis uses publicly available bridge inspection and transportation datasets from the Federal Highway Administration (2025) and summary reports from the Nebraska - Department of Transportation (2025). These datasets contain information on bridge conditions, construction characteristics, and traffic volumes for bridges across the United States. The Nebraska bridge inventory dataset for 2025 was used to evaluate bridge conditions in **Douglas County**, which includes the city of **Omaha**. The purpose of the analysis is to estimate the potential risk of bridge collapse by examining structural condition ratings and transportation usage.

3.3.1 Data Sources

- **National Bridge Inventory (NBI)**: A comprehensive dataset maintained by the Federal Highway Administration (2025) that includes inspection results, structural condition ratings, bridge dimensions, construction year, and traffic statistics for bridges in the United States.¹

¹The Nebraska NBI extract was downloaded from the FHWA bridge data portal at <https://www.fhwa.dot.gov/bridge/nbi/ascii.cfm>.

- **Nebraska Department of Transportation Bridge Reports:** Provides summary information on bridge conditions across Nebraska, including counts of structurally deficient or functionally obsolete bridges and maintenance planning data (Nebraska - Department of Transportation 2025). ²

3.3.2 Data Features

The datasets include several variables relevant to evaluating bridge safety and infrastructure risk:

- **Structural Condition Ratings:** Includes inspection scores for deck condition `DECK_COND_058`, superstructure condition `SUPERSTRUCTURE_COND_059`, and substructure condition `SUBSTRUCTURE_COND_060`. These ratings range from 0 to 9, where higher values represent better structural condition.
- **Traffic Volume:** Average Daily Traffic `ADT_029` estimates the number of vehicles crossing each bridge per day, indicating the level of transportation demand.
- **Bridge Age:** Construction year `YEAR_BUILT_027` allows calculation of bridge age, which may influence structural deterioration over time.
- **Location Information:** Geographic identifiers such as state code and county code allow bridges in Douglas County to be isolated for analysis.
- **Bridge Dimensions and Characteristics:** Additional variables such as bridge length, roadway width, and number of spans describe the structural characteristics of each bridge.

3.3.2.1 Missing Data

Prior to filtering to Douglas County, a missing data assessment was conducted on the key structural condition variables. Missingness was observed primarily in the three condition rating fields (`DECK_COND_58`, `SUPERSTRUCTURE_COND_059`, `SUBSTRUCTURE_COND_060`) and in `ADT_029`. Bridges with missing values in any of these variables were excluded from analysis using `drop_na()`, as these fields are essential for risk classification. It is worth noting that missingness in condition ratings may not be random, smaller or lower-priority structures are sometimes exempt from full NBI inspection requirements, which could mean the excluded bridges are systematically different from those retained. This is a limitation of the analysis.

²Reports are publicly available at <https://dot.nebraska.gov/media/4efjqkfa/bridge-condition-locally-owned.pdf>.

3.3.3 Data Collection and Preparation

The Nebraska National Bridge Inventory dataset was downloaded in delimited text format and converted into a CSV file for analysis. The dataset was then imported into the R programming environment. Package versions and system information are documented in Section 6.1.

To focus on the study area, bridges located in Douglas County were selected using the county code variable (`COUNTY_CODE_003 == 55`). Data cleaning steps were performed to prepare the dataset for statistical analysis. Several inspection variables were initially stored as character values and were converted to numeric format in order to calculate summary statistics. Missing or non-numeric values were treated as missing observations and excluded.

Additional variables were created to support analysis. Bridge age was calculated by subtracting the construction year from the most current year (2025). This variable helps identify older infrastructure that may require increased maintenance or monitoring.

3.3.3.1 Data Classification

To evaluate structural risk, bridges were categorized according to their inspection ratings.

- **Higher-risk bridges:** Bridges with structural condition ratings of 4 or lower in the deck, superstructure, or substructure categories, indicating poor structural condition or advanced deterioration.
- **Moderate-condition bridges:** Bridges with ratings between 5 and 6, indicating fair structural condition.
- **Good-condition bridges:** Bridges with ratings of 7 or higher, representing satisfactory or good structural performance.

This classification aligns with FHWA conventions for identifying structurally deficient bridges and allows the analysis to prioritize structure most in need of intervention.

3.3.3.2 R packages and Software

The analysis was conducted in R. Package versions are recorded in Section 6.1. Key packages used include:

- **readr:** Used to import the bridge inventory dataset into R (Wickham, Hester, et al. 2026).
- **tidyverse:** Used for data manipulation, filtering, and summarization (Wickham and RStudio 2023).

- **dplyr:** Used to calculate descriptive statistics and identify high-risk bridges (Wickham, François, et al. 2026).
- **ggplot2:** Used to create visualizations of bridge condition ratings, traffic levels, and bridge age distributions (Wickham, Chang, et al. 2026).
- **knitr:** Used to generate the final report and integrate code with written analysis (“Knitr - Yihui Xie | ” 2026).
- **scales:** Used for graphical scales map data to aesthetics (Wickham et al. 2025).

3.3.4 Analytical Approach

The analysis proceeded in three stages. First, descriptive statistics were calculated to characterize the full sample of Douglas County bridges on key variables: structural condition ratings, bridge age, and traffic volume. Second, a condition rating analysis identified bridges with ratings at or below 4 in any structural component, flagging them as higher-risk. Third, a combined risk assessment cross-tabulated structural condition against traffic exposure (ADT), since bridges that are both structurally compromised and heavily used represent the greatest public safety risk — high likelihood of failure combined with high consequence. Spearman correlation tests were used to assess the monotonic relationships between age, condition, and traffic, given the ordinal nature of the condition rating variables.

4 Result

4.1 Study Sample Characteristics

Bridge inspection data from the Federal Highway Administration were analyzed for bridges located in Douglas County, which includes the city of Omaha. A total of 390 bridges were included in the analysis after excluding records with missing values in key structural or traffic variables.

Table 4.1: Descriptive statistics for traffic volume and bridge age among analyzed Douglas County bridges

Variable	Min	Median	Mean	Max
Daily Traffic Mean	25	15,750	36,947	182,320
Bridge Age (years)	2	39	41	106

As shown in Table 4.1, Average Daily Traffic ranged from 25 to 182,320 vehicles per day, with a median of 15,750 and a mean of 36,947. This wide spread reflects the mix of major arterial crossings and minor local bridges in Douglas County

Bridge age ranged from 2 to 106 years, with a median of 39 years and a mean of 41.3 years. The relatively high mean age suggests that a substantial portion of the county’s bridge inventory predates modern design standards and may face greater maintenance demands.

The mean deck, superstructure, and substructure ratings were 6.73, 7.02, and 6.95, respectively, all falling within the “fair” range (5-6) on the 0-9 NBI scale, while the superstructure mean marginally exceeds the good threshold (≥ 7), indicating the overall structural inventory is in fair-to good condition.

4.2 Structural Condition

Structural conditions were evaluated using inspection ratings for bridge decks, superstructures, and substructures. These ratings range from 0 (failed condition) to 9 (excellent condition). Since these ratings are recorded as discrete integers rather than continuous measurements,

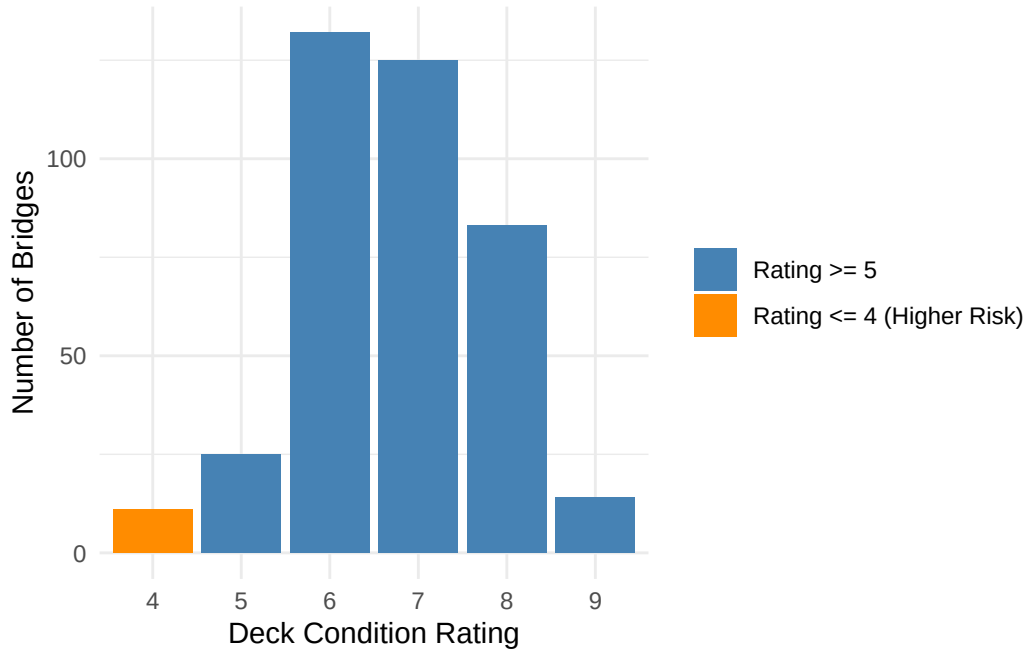


Figure 4.1: Distribution of Bridge Deck Condition Ratings in Douglas County, Nebraska.

they are best treated as ordered categorical variables; Figure 4.1 therefore displays counts as a bar chart rather than a histogram.

The majority of bridges received deck condition ratings of 5, 6, or 7, consistent with the sample mean of 6.73. However, a meaningful minority of bridges fall at or below the rating-4 threshold. A Spearman correlation test confirmed a statistically significant negative relationship between bridge age and deck condition ($\rho = -0.4$, $p = 0$), indicating that older bridges tend to exhibit lower deck condition ratings — though the effect is modest, suggesting factors beyond age alone (e.g., maintenance history, traffic loading, construction materials) also play a role.

4.3 Traffic Exposure

Traffic exposure was analyzed to evaluate the potential *consequences* of bridge failure. A structurally compromised bridge that carries tens of thousands of vehicles per day poses a fundamentally different public risk than one serving a low-volume rural road.

As shown in Figure 4.2, the distribution is left-skewed even on a log scale, with a small number of bridges supporting extremely high traffic volumes (exceeding 100,000 vehicles per day). These high-volume bridges represent critical nodes in Omaha's transportation network where a failure would have the most immediate regional impact.

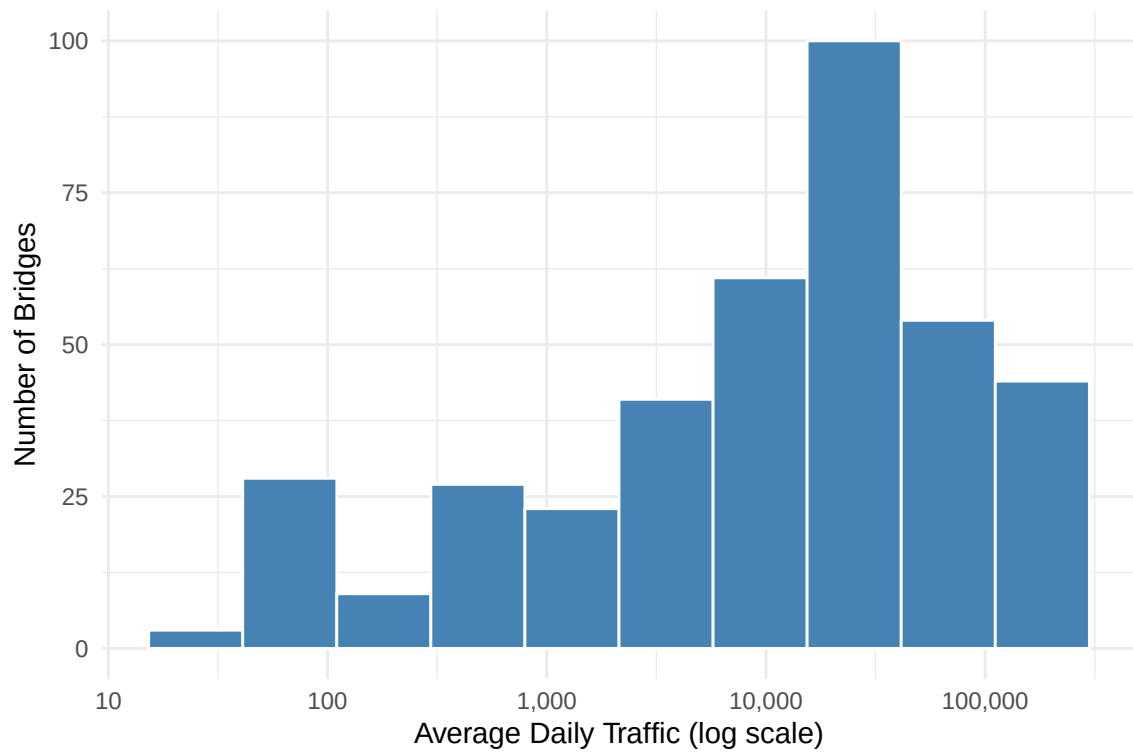


Figure 4.2: Distribution of Average Daily Traffic (ADT) for bridges in Douglas County, plotted on a logarithmic scale to accommodate the wide range of values. Most bridges carry between 1,000 and 100,000 vehicles per day.

The correlation between deck condition and traffic volume was weak and not statistically significant ($\rho = 0.17$, $p = 0.001$). This is an important finding: bridge condition does not appear to be better or worse on higher-traffic routes, which means high-traffic bridges are not receiving systematically stronger maintenance relative to their exposure.

Of the 390 bridges analyzed, 16 (4.1%) were classified as higher-risk based on having at least one structural condition rating at or below 4. The remaining 374 bridges were classified as lower risk.

Higher-risk bridges were, on average, 29.7 years older than lower-risk bridges (mean age 69.8 vs. 40 years), reinforcing the age-condition relationship observed above. Their mean daily traffic (20,076 vehicles/day) was lower than that of lower-risk bridges (37,669 vehicles/day), though the difference should be interpreted cautiously given the high variability in ADT across both groups.

4.4 Traffic by Bridge Condition

The combined risk picture — structural condition against traffic load — is the central question for prioritizing interventions. Figure 4.3 displays the distribution of ADT by deck condition rating.

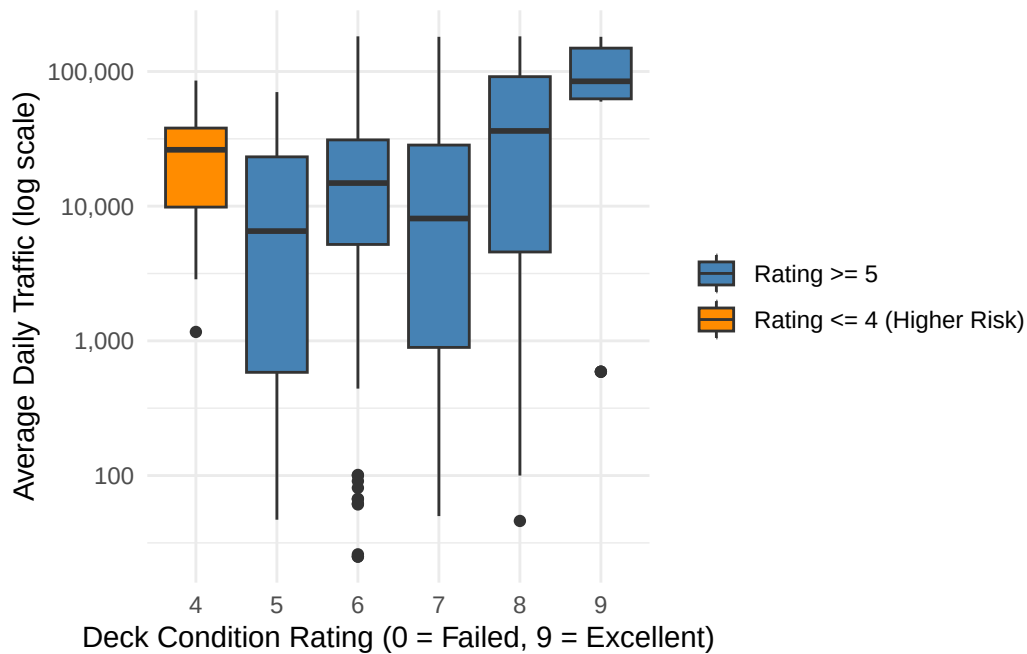


Figure 4.3: Relationship between bridge structural condition and traffic exposure in Douglas County, Nebraska.

Figure 4.3 reveals a finding that carries direct policy implications: several bridges rated at 4 or below carry daily traffic volumes in the tens of thousands. These bridges sit in the upper-left danger zone — structurally compromised yet heavily used — and represent the clearest candidates for prioritized inspection and intervention. At the same time, the wide interquartile ranges within each rating category confirm that condition rating alone is not a reliable proxy for traffic exposure; both dimensions must be considered together when allocating limited maintenance resources.

5 Conclusion

This analysis examined bridge risk in Omaha using a framework that combines structural condition and traffic exposure. Of the 390 bridges analyzed, only 16 were classified as high risk, indicating that overall risk is concentrated in a small subset of structures. High-risk bridges were substantially older (mean age 69.8 years vs. 40.0 years), and age was moderately associated with poorer condition ($\rho = -0.40$, $p < 0.001$). However, traffic volume showed only a weak relationship with condition ($\rho = 0.17$, $p = 0.001$), and high-risk bridges carried lower average traffic than lower-risk bridges. These findings suggest that structural deterioration and usage do not always align, reinforcing the need to evaluate both likelihood and consequence when assessing bridge risk.

While informative, this study has several limitations. The analysis relies on publicly available data from the Federal Highway Administration (2025), which may include inconsistencies or missing values. In addition, proxy variables such as condition ratings and traffic volume do not fully capture true failure probability or real-world impact. The focus on a single region limits generalizability, and the absence of observed failure events reduces statistical power and limits predictive modeling.

5.1 Recommendation

Based on the findings, the following are recommended:

! Priority Action: Targeted Intervention for High-Risk Bridges

Only 16 of 390 bridges were classified as high risk. Resources should be concentrated on this small subset immediately — these bridges are, on average, nearly 30 years older than lower-risk structures and show clear signs of advanced deterioration. Delaying intervention on this group represents the greatest near-term safety liability.

💡 Use a Two-Dimensional Risk Framework

Traffic volume is best understood as a measure of **economic consequence**, not a causal factor in structural deterioration. A bridge in poor condition on a low-traffic road may be structurally unsafe but pose limited disruption if it fails; the same condition rating on

a high-traffic corridor represents a much higher societal cost. Maintenance prioritization should therefore weigh structural condition (likelihood of failure) and traffic exposure (consequence of failure) separately, rather than treating either dimension alone as a proxy for overall risk.

i Improve Data Collection for Lower-Traffic Structures

The weak relationship between traffic volume and condition ratings may partly reflect gaps in inspection data for smaller or lower-priority bridges. Standardizing inspection frequency and completeness across all bridge types would improve the reliability of future risk assessments.

References

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6 Session Information

6.1 Session Information

The following records the R session information used to produce this analysis, including package versions and system details.

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.6.1
```

```
Matrix products: default
BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/Libraries/libBLAS.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
```

```
locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: America/Chicago
tzcode source: internal
```

```
attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base
```

```
other attached packages:
[1] sessioninfo_1.2.3
```

```
loaded via a namespace (and not attached):
[1] compiler_4.5.2    fastmap_1.2.0     cli_3.6.5         tools_4.5.2
[5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.17.1 yaml_2.3.12
[9] rmarkdown_2.30    knitr_1.51        jsonlite_2.0.0    xfun_0.55
[13] digest_0.6.39     rlang_1.1.7       evaluate_1.0.5
```